the iron-rich layers always occur below the manganese-rich layers.

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# Additional Paleomagnetic Data on the Baffin Diabase Dikes and a Revised Franklin Pole

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Paleomagnetic data were obtained from eleven additional sites on Baffin diabase dike intrusions (part of the Franklin diabases). The rock at four sites was found to be normally magnetized, at six sites reversely magnetized, and at one site the rock contained no primary remanent magnetization that could be isolated by alternating field (a.f.) demagnetization. Baffin Island is divisible into several zones within which the Baffin dikes are either normally or reversely magnetized. This may indicate that more than one reversal is represented. The ten sites yield a pole at 168° E, 6° N, $\alpha_{95}$ = 5°, and when combined with previously published data (total 46 sites) yield a revised Franklin diabase pole at 166° E, 6° N, $\alpha_{95}$ = 4°. At least 10 of the 11 new sites apparently lie outside the zone from within which Baffin dikes have yielded anomalous

remanent magnetization directions. Thermomagnetic curves for representative material of the Baffin dikes indicate that the magnetic mineral of these rocks is almost pure magnetite. It does not explain the anomalous magnetization of the Franklin dikes that occur along the northeast coastal area.

Des données sur le paléomagnétisme ont été obtenues pour onze autres sites sur les intrusions du dyke de diabase de Baffin (partie des diabases de Franklin). A quatre de ces sites, les roches sont normalement magnétisées, à six autres, elles montrent un magnétisme inverse, et à un endroit la démagnétisation a.f. n'a décelé aucun magnétisme rémanent primaire dans la roche. L'île Baffin peut être divisée en plusieurs régions à l'intérieur desquelles les dykes de Baffin sont ou bien normalement ou bien inversement magnétisés, ce qui pourrait témoigner de plus d'une inversion. Les dix sites donnent un pôle à 168° E, 6° N, $\alpha_{95} = 5°$, et si l'on tient compte de données déjà publiées (au total 46 sites), on obtient un pôle à 166° E, 6° N, $\alpha_{95} = 4°$. Au moins 10 de ces 11 nouveaux sites se situent apparemment à l'extérieur de la zone où les dykes de Baffin ont montré des directions anormales de magnétisation rémanente. Des courbes thermomagnétiques pour des matériaux représentatifs des dykes de Baffin indiquent que le minéral magnétique de ces roches est de la magnétite presque pure. La magnétisation anormale des dykes de Franklin le long de la région côtière nord-est reste inexpliquée. [Traduit par le journal]

## Introduction

Franklin diabasic intrusions of Hadrynian age (K–Ar age 675 m.y.) occur in a broad belt across the northern Canadian shield (Fahrig *et al.* 1971, Robertson and Baragar 1972) and possibly elsewhere in the shield (Murthy 1971). The Natkusiak volcanics of Victoria Island are probably extrusive equivalents of the diabasic dike and sheet intrusions. Franklin intrusive rocks have been used to define the Franklin Magnetic Interval (Fahrig *et al.* 1971) which is that period of time during the formation of Franklin intrusive and extrusive rocks when the earth's field had a configuration such that rocks from the northern Canadian Shield define a pole at 167° E, 8° N ($\alpha_{95} = 5°$) and during which period there was at least one polarity reversal. It is emphasized here that magnetic time intervals are defined on a basis of the rocks and their paleomagnetic properties. The radioisotopic age of the Franklin rocks plays no part in this definition. Franklin intrusions (Baffin dikes) occur throughout Baffin Island and in an earlier study it was found that the dikes along the most northeasterly coastal area exhibit a magnetization that is anomalous relative to that of the remaining Franklin rocks. These anomalously magnetized dikes yield a pattern of poles that is streaked toward the present magnetic pole. Although there is some overlap, the rocks of these dikes have lower coercivities as indicated by the median destructive fields of their natural (n.r.m.), anhysteretic (a.r.m.), and saturation isothermal (i.r.m.) remanent magnetization. It was suggested that this zone was exposed to mild reheating during the most recent (Mesozoic) opening of Davis Strait (Fahrig *et al.* 1971). It seemed of special interest to determine whether Baffin dikes displaying these characteristics are indeed restricted to the northeast coastal area of the island, to collect further data on the distribution of normally and reversely polarized dikes of Baffin Island, and to collect more data for the corroboration or revision of the Franklin pole.

## Paleomagnetic Method

A total of 67 oriented drill cores, 2.5 cm in diameter, were collected from 11 sites. Orientation was by sun compass and the measurement of natural remanent magnetization (n.r.m.) was carried out with an automated biastatic magnetometer (Larochelle and Christie 1967). Two specimens were cut from each core and in homogeneously magnetized cores the average value of the two specimens was used as basic datum.

Elimination of unwanted secondary components was achieved by carrying out alternating field (a.f.) demagnetization. The optimum field was chosen by standard method (As and Zijderveld 1958, Irving *et al.* 1961) in which 2 specimens from each site were step cleaned. The magnetization directions of all the test specimens with the exception of those from site 7 moved directly away from the present earth's field direction and exhibited excellent end points so the remaining specimens of the 10 sites were demagnetized in optimum peak fields that ranged from 200 to 300 Oe (Table 1, Fig. 2). The n.r.m. directions of the specimens from site 7 were closely grouped in the direction of the present earth's field. The step cleaning of two specimens from this site did

TABLE 1. Directions of remanent magnetization before (n.r.m.) and after magnetic cleaning. Pole positions of Baffin dikes and revised Franklin pole position

| | | | n.r.m. | | | | | | a.f. | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site | $N$ | $n$ | $D$ | $I$ | $\delta°$ | $k$ | | | | $D$ | $I$ | $\delta°$ | $k$ | $\delta_{\bar{R}}°$ |
| 1 | 6 | 6 | 110 | 50 | 11 | 55 | | | 250 | 123 | −1 | 7.8 | 108 | 3.2 |
| 5 | 6 | 6 | 122 | 44 | 20 | 16 | | | 200 | 127 | 0 | 3.8 | 446 | 1.6 |
| 6 | 5 | 5 | 116 | 46 | 21 | 14 | | | 200 | 125 | 16 | 16 | 24 | 7.5 |
| 8 | 7 | 6 | 93 | 75 | 8.4 | 93 | | | 200 | 126 | 20 | 9.2 | 78 | 3.8 |
| 39 | 7 | 4 | 114 | 40 | 7.4 | 121 | | | 300 | 116 | 7 | 7.4 | 119 | 3.7 |
| 40 | 6 | 5 | 105 | 73 | 3.8 | 461 | | | 250 | 122 | 31 | 11 | 54 | 4.9 |
| Mean (south) pole position from 6 cleaned Baffin sites listed above[1] | | | | | | 168.3° E | 6.0° N | $k = 64$ | $\alpha = 7.2°$ | | | | | |
| 2 | 6 | 4 | 306 | 58 | 10 | 61 | | | 250 | 304 | −9 | 14 | 35 | 6.9 |
| 3 | 6 | 5 | 306 | 32 | 22 | 14 | | | 250 | 300 | −18 | 11 | 54 | 4.9 |
| 4 | 6 | 4 | 321 | 69 | 14 | 31 | | | 200 | 311 | −4 | 8.5 | 92 | 4.2 |
| 38 | 7 | 4 | 314 | 41 | 15 | 28 | | | 200 | 306 | −17 | 9.9 | 66 | 5.0 |
| Mean (north) pole position from 4 cleaned Baffin sites listed above | | | | | | 167.5° E | 6.6° N | $k = 179$ | $\alpha = 5.2°$ | | | | | |
| Combined (north and south poles) for 10 Baffin sites listed above | | | | | | 168.2° E | 6.3° N | $k = 84$ | $\alpha = 4.8°$ | | | | | |
| Revised Franklin pole $n = 46$[2] | | | | | | 165.8° E | 5.8° N | $k = 29$ | $\alpha = 3.8°$ | | | | | |

NOTES: $N$ = number of samples collected at the site; $n$ = number of samples left after eliminating those cores which are inhomogeneously magnetized (θ greater than 20°); $D$ = declination; $I$ = inclination; $\delta°$ = angular standard deviation; a.f. = peak alternating field in oersteds; $k$ = Fisher's estimate of precision; $\alpha_{95}$ = half angle of the cone of confidence at $P = 0.95$; $\delta_{\bar{R}}°$ = standard error of the mean.

[1]All mean poles were calculated from site pole positions.

[2]Revised Franklin pole combined above data (10 sites); with those of Robertson and Baragar (1972) (13 sites), and the Group A data (23 sites) of Fahrig *et al.* (1971).

not follow a simple pattern. One specimen exhibited large random directional changes after cleaning in a.c. fields above 50 Oe and the other in fields above 150 Oe. All the specimens of this site were then cleaned in a series of steps. The magnetization direction of individual cores shifted away from present earth's field direction but in an apparently random fashion and the angle between the two specimens of each core increased. The grouping of within-site directions became steadily more random. As the original n.r.m. site direction and that remaining after cleaning at low fields was close to the present earth's field direction and quite different from cleaned directions of other Baffin dikes (either normal or reversed) it is extremely unlikely that this is the primary thermal remanent magnetization (t.r.m.) direction of site 7. For this reason data for the 5 cores of this site were not used in the group statistics. The angle, ($\theta$), between specimens of the same core in 13 of the other 63 cores (21%) was greater than 20° after cleaning at the optimum fields for the sites to which they belong, so data from these inhomogeneously magnetized cores were also excluded from the statistics.

## Paleomagnetic Results

The n.r.m. directions formed an almost perfect girdle (Fig. 2), none of the sites was internally random (Table 1), and after optimum a.f. cleaning, sample directions fell into two groups whose directions were antiparallel and gently inclined in the southeast and northwest quadrants. The rock at four of the sites (the northwesterly magnetized sites) exhibits normal polarity and at the remaining 6 sites (the southeasterly magnetized sites) reversed polarity. The antiparallelism of the two groups is illustrated by the excellent agreement in mean pole positions of the normal and reversed groups. These differ by less than one degree of arc (Table 1). Figure 1 shows the aerial distribution of normally and reversely polarized Baffin dikes (25 sites) as indicated by the present, and previous studies (Fahrig *et al.* 1971). This indicates that there are at least two zones of reversed and one of normal polarity within Baffin Island. These discrete zones of perfectly reversed dikes are rather strong evidence that we are indeed dealing with a successfully isolated primary thermal remanent magnetization (t.r.m.). It is expected that

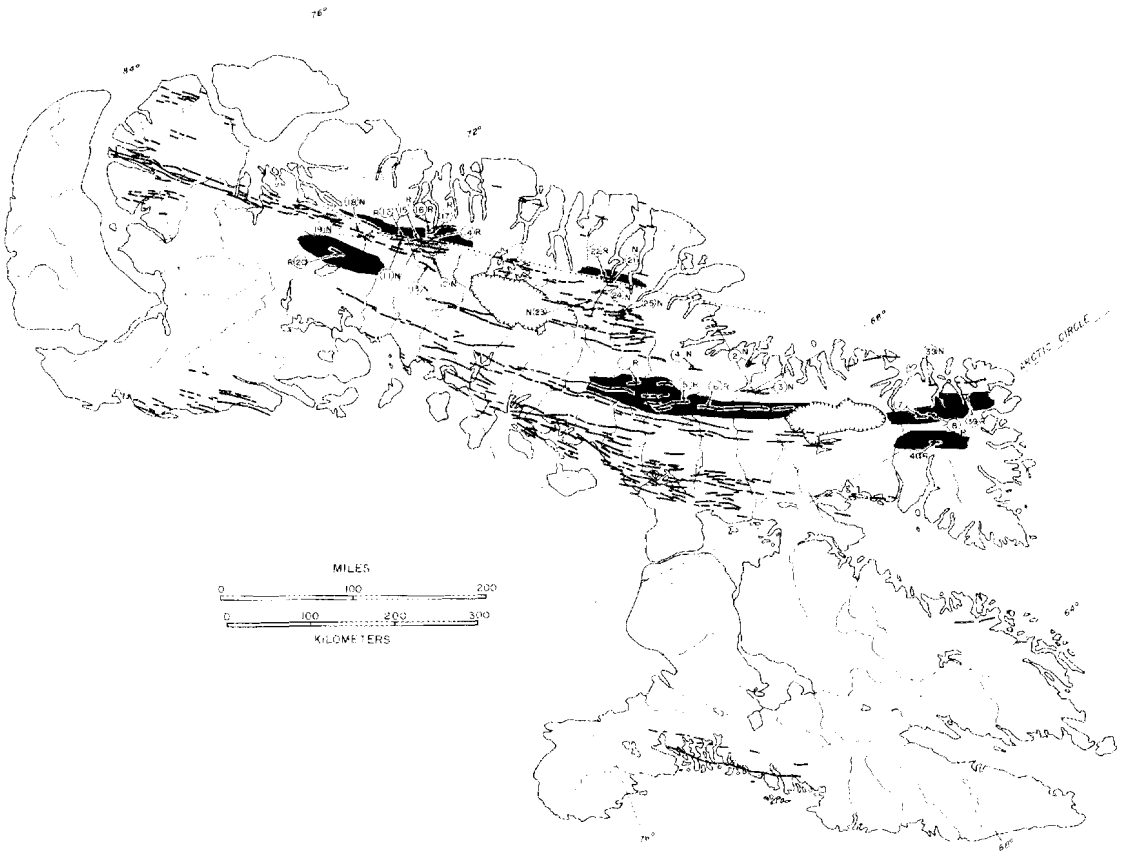



FIG. 1. Baffin dikes and sampling sites. Data for sites 1 to 8 and 38 to 40 are reported in the present paper. Data from other sites are reported in Fahrig *et al.* (1971). Zones of dikes (sites) of normal polarity are shown stippled, those of reversed polarity, solid black. Sites northeast of the dotted line exhibit anomalous magnetization.

future work will further define the distribution of normal and reversed Franklin dike zones in this part of the shield.

The present results (10 sites) when combined with previous results (Fahrig *et al.* 1971, Robertson and Baragar 1972) have been used to compute a very reliable revised Franklin pole at 166 °E, 6 °N, $\alpha_{95} = 4°$, $N = 46$. Details are given in Table 1.

### Curie Temperature and Coercive Force

It was thought desirable to study further the properties of the magnetic minerals of representative material from both the present collection (11 sites) and the previous collection (15 sites) from the dikes of Baffin Island. The major objective was to investigate the anomalous remanent magnetization of six dikes which occur along the northeastern coast of Baffin (Fahrig *et al.* 1971) and in particular the suggestion that the magnetic minerals in this zone underwent significant modifications in response to mild heating that accompanied the Mesozoic opening of Davis Strait.

Thermomagnetic curves were obtained on 50 mg pieces using a recording magnetic force balance (Schwarz 1968). The dikes examined were 3 previously reported 'normal' and 6 anomalous dikes (Groups A and B of Fahrig *et al.* 1971) and one dike from the present group. All of the curves are simple and indicate the presence of a single magnetic mineral. The six anomalous dikes had magnetic minerals with Curie temperatures in the range 525 to

TABLE 2. Intensities of natural (n.r.m.) magnetization in c.g.s. units/cm$^3$, the median destructive field (m.d.f., n.r.m.) of the natural magnetization; the median destructive field (m.d.f., i.r.m.) of the saturation isothermal magnetization, ferrous-ferric ratios

| Site | n.r.m. $\times 10^{-4}$ | m.d.f. n.r.m. | m.d.f. i.r.m. | FeO $Fe_2O_3$ | Polarity |
|---|---|---|---|---|---|
| 1 | 5.25 | 175 | 336 | 3.5 | R |
| 2 | 2.57 | 185 | 164 | 2.7 | N |
| 3 | 5.30 | 190 | 275 | 4.0 | N |
| 4 | 4.97 | 140 | 154 | 2.6 | N |
| 5 | 5.32 | 260 | 296 | 3.3 | R |
| 6 | 10.00 | 170 | 342 | 3.0 | R |
| 7 | 8.71 | 130 | 382 | 2.8 | R |
| 8 | 6.58 | 155 | 378 | 3.1 | R |
| 38 | 6.19 | 210 | 347 | 3.1 | N |
| 39 | 6.41 | 215 | 287 | 3.4 | R |
| 40 | 11.25 | 160 | 445 | 2.4 | R |
| Average* | 5.51 | 185 | 300 | 3.1 | |

*The average values do not include those for the specimen from site 7.

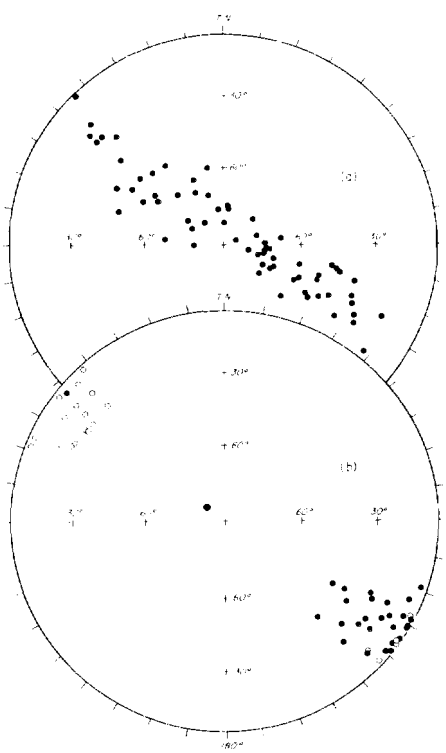


FIG. 2. Natural remanent magnetization directions (solid circles are north-seeking down) of 11 sites before a.f. cleaning (a) and after optimum a.f. cleaning (b). The latter does not include 13 samples whose $\theta$ exceeds 20° or the samples from site 7. Present earth's field direction at the collection site is indicated by larger dot.

560° with an average of 550°. The four 'normal' dikes contain magnetic minerals with a range of 540 to 565° and with an average of 555°. These data suggest that the magnetic mineral is almost pure magnetite. Furthermore as no consistent difference was observed between the curves for anomalously magnetized and 'normally' magnetized dikes there is no significant difference in the magnetic mineralogy between the two groups.

The coercive force of the saturation remanent magnetization of the 11 new sites was determined by measuring the magnetization of one specimen from each site after magnetization in a constant 8000 Oe field and the application of a step-wise increasing backfield. The coercive force as indicated by the median destructive field of this i.r.m. is given in Table 2. This force varies considerably as observed in the earlier study (Fahrig *et al.* 1971, Group A results) and has an average (300) which is similar to that obtained in the earlier study (317) (Fahrig *et al.* 1971).

The median destructive field of the n.r.m. obtained from the a.f. intensity curves for one specimen from each of the 11 new sites is also given in Table 2. The average value (185) is significantly higher than in earlier work (Group A results, 124, Fahrig *et al.* 1971). The specimen from site 7 exhibited the lowest coercivity (130) and, as has been noted this site did not yield reliable n.r.m. data. On the other hand,

the other anomalous samples tend to yield low coercivities for all the types of remanence tested. In the absence of a mineralogical difference, it appears that the anomalous behavior is caused by a consistent difference in size of the magnetite grains in the range of a few hundred angstroms to a few hundred micrometers. It is also possible that "magnetic softening" took place in the anomalous dikes during an extended period of mild thermal metamorphism.

## Discussion

The apparent uniformity of the magnetic mineralogy of the Baffin dikes as indicated by their thermomagnetic curves and Curie temperatures leaves the problem of the anomalously magnetized dikes unresolved. The mechanism of their magnetization may be a complex one, involving intrusion and consolidation during a field reversal and later remagnetization accompanying subtle mineralogical changes resulting from mild reheating during the Mesozoic opening of Davis Strait. In this regard Murthy's (1971) work on the Grenville dikes is of special interest. These dikes are also Late Proterozoic (Hadrynian) in age, and those sites that appear to exhibit an initial stable t.r.m. yielded a pole position close to that of the Franklin. There is good reason then to believe that the Grenville dikes are also members of the Franklin diabase intrusions. Murthy was able to correlate instability with titanomaghemite content, the dikes exhibiting more titanomaghemite being least stable. There is also the remarkable coincidence that the Grenville sites (only 1/3 of which exhibited stable t.r.m.) are either from within, or immediately adjacent to the Ottawa–Bonnechere graben of the St. Lawrence rift system (Kumarapeli and Saull 1966). Although rifting in this region may have started in Late Precambrian time and may have been responsible for Grenville dike intrusion, movement has certainly taken place in Mesozoic to Recent time. Here as in Baffin Island we have Hadrynian dike intrusions, some of which may have been mildly reheated during Mesozoic to Recent rifting. It is extremely tempting to tie the magnetization of both the anomalous Baffin and anomalous Grenville dikes to these tectonic events.

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